

```

#include <iostream>
#include <string>
using namespace std;

int main() {
    int roll[100], n, choice, pos, r;
    string name[100];
    float cgpa[100];

    cout << "Enter number of students: ";
    cin >> n;

    // Input student details
    for (int i = 0; i < n; i++) {
        cout << "\nEnter details " << i + 1 << endl;
        cout << "Roll No: ";
        cin >> roll[i];
        cout << "Name: ";
        cin >> name[i];
        cout << "CGPA: ";
        cin >> cgpa[i];
    }

    do {
        cout << "\n----- Student Information System -----" <<
endl;
        cout << "1. Insert Student" << endl;
        cout << "2. Delete Student" << endl;
        cout << "3. Search Student" << endl;
        cout << "4. Display Students" << endl;
        cout << "5. Exit" << endl;
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {

        case 1:
            cout << "Enter position (1 to " << n + 1 << "): ";
            cin >> pos;

            if (pos < 1 || pos > n + 1) {
                cout << "Invalid Position" << endl;
                break;
            }

            for (int i = n; i >= pos; i--) {
                roll[i] = roll[i - 1];
                name[i] = name[i - 1];
                cgpa[i] = cgpa[i - 1];
            }

            cout << "Enter Roll No: ";
            cin >> roll[pos - 1];
            cout << "Enter Name: ";

```

```

        cin >> name[pos - 1];
        cout << "Enter CGPA: ";
        cin >> cgpa[pos - 1];

        n++;
        cout << "Student Inserted." << endl;
        break;

    case 2:
        cout << "Enter Roll No to Delete: ";
        cin >> r;

        for (int i = 0; i < n; i++) {
            if (roll[i] == r) {
                for (int j = i; j < n - 1; j++) {
                    roll[j] = roll[j + 1];
                    name[j] = name[j + 1];
                    cgpa[j] = cgpa[j + 1];
                }
                n--;
                cout << "Student Deleted." << endl;
                break;
            }
        }
        break;

    case 3:
        cout << "Enter Roll No to Search: ";
        cin >> r;

        for (int i = 0; i < n; i++) {
            if (roll[i] == r) {
                cout << "\nStudent Found" << endl;
                cout << "Roll No: " << roll[i] << endl;
                cout << "Name: " << name[i] << endl;
                cout << "CGPA: " << cgpa[i] << endl;
                break;
            }
        }
        break;

    case 4:
        cout << "\nStudent Records" << endl;
        cout << "Roll No\tName\tCGPA" << endl;

        for (int i = 0; i < n; i++) {
            cout << roll[i] << "\t" << name[i] << "\t" <<
cgpa[i] << endl;
        }
        break;

    case 5:
        cout << "Program Ended..." << endl;
        break;

```

```
        default:
            cout << "Invalid Choice!" << endl;
        }

    } while (choice != 5);

    return 0;
}
```